

VCF Private Cloud Deployment – Implementation Procedure

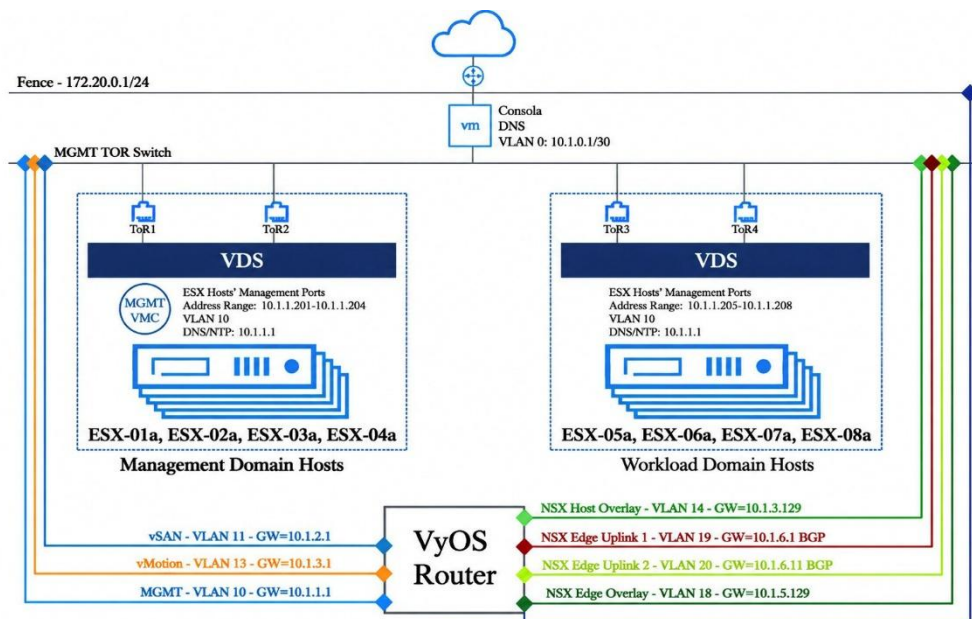
This document describes the implementation of a VCF Private Cloud environment, from the initial network design through the automated deployment of the VCF platform. The lab topology uses redundant Top-of-Rack (ToR) connectivity together with a VyOS router to provide the required Layer 3 gateways and routing.

The environment is divided into a Management Domain and a Workload Domain. The Management Domain uses four ESXi hosts (ESX-01a through ESX-04a), while the Workload Domain contains four additional hosts (ESX-05a through ESX-08a). Virtual Distributed Switches (VDS) are used to provide consistent connectivity across the hosts.

Traffic is separated by function using dedicated VLANs for management, vSAN, vMotion, and NSX traffic. The VyOS router provides the corresponding default gateways and supports the routing required by the NSX Edge connectivity, including BGP sessions.

A dedicated DNS/Console VM supports name resolution and administrative access. The management segment is VLAN 10 (10.1.1.0/24), vSAN uses VLAN 11, vMotion uses VLAN 13, and NSX overlay and Edge traffic use separate network segments.

Together, this design establishes the addressing, segmentation, routing, and connectivity prerequisites for the VCF Private Cloud implementation.



VCF Private Cloud – Logical Design

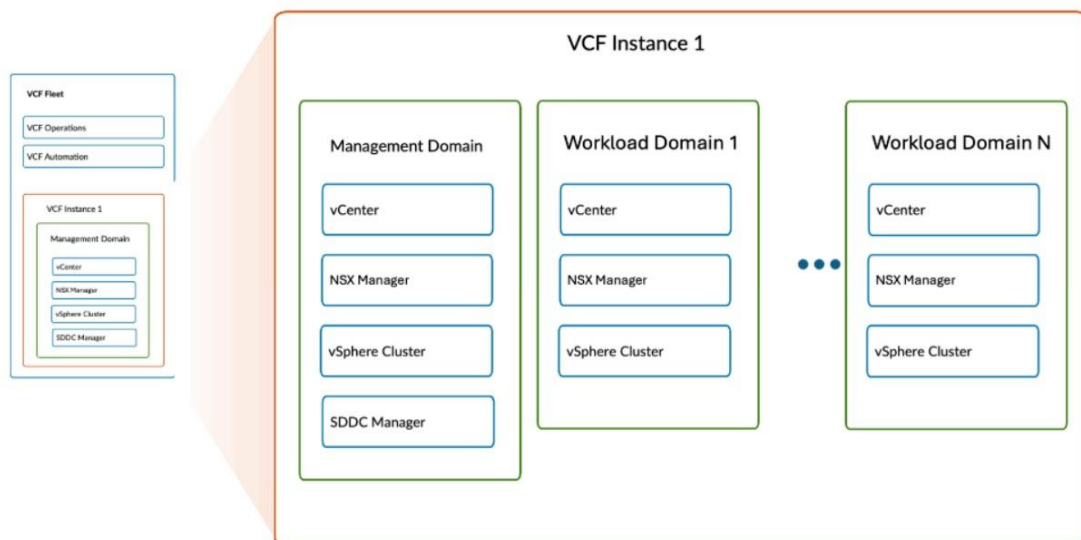
The platform is structured as a VCF Instance containing a Management Domain and one or more Workload Domains.

The Management Domain contains the core platform services, including vCenter, NSX Manager, the vSphere cluster, and SDDC Manager. Workload Domains are separated environments intended for application or tenant workloads and include their own vCenter, NSX Manager, and vSphere cluster.

VCF Fleet provides the higher-level management layer, with VCF Operations and VCF Automation used to support centralized operations, automation, and lifecycle management across the VCF environment.

Full Image

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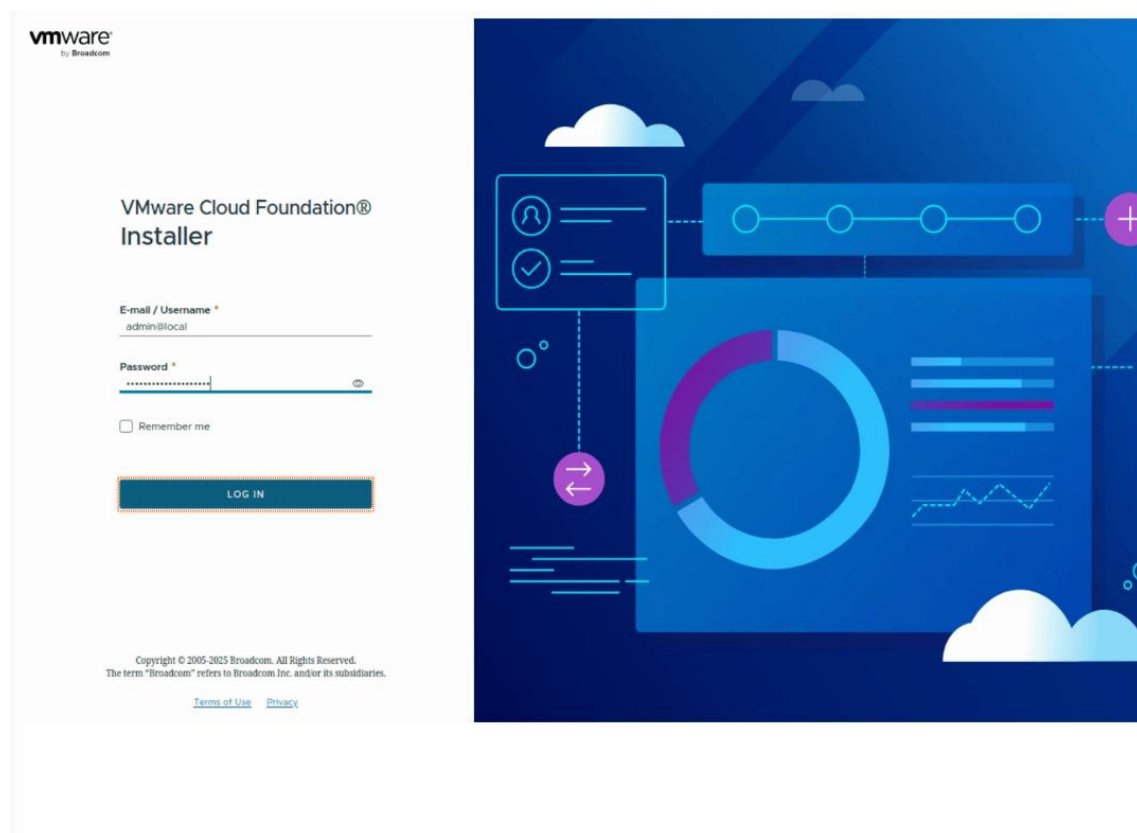
VCF Installer – Initial Configuration

Accessing the VCF Installer

After the VCF Installer appliance has been deployed, open its assigned IP address in a web browser. The initial login uses the local administrative account shown by the interface, typically 'admin@local'.

Log in to the installer:

1. Enter the configured administrator password.
2. Select 'LOG IN' to access the VCF Installer.

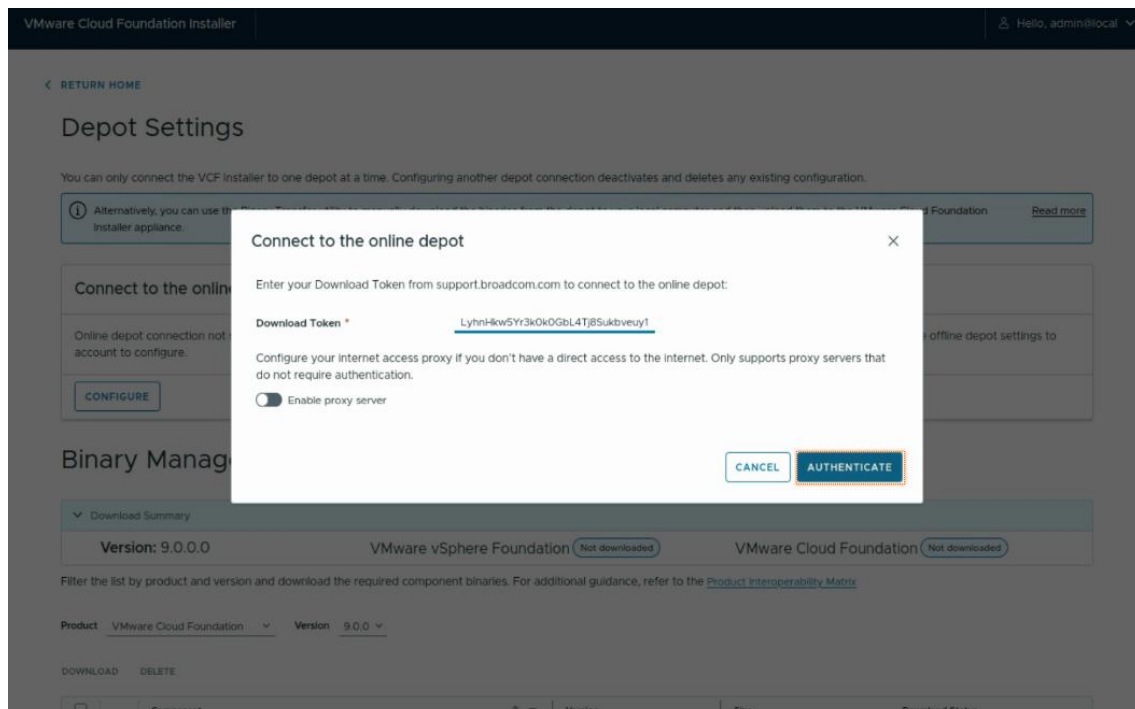


Software Depot Configuration

Before starting the deployment, configure access to the software depot so that the installer can obtain the binaries required for the selected VCF release. For an online connection, a Broadcom download token is required.

1. Open 'DEPOT SETTINGS AND BINARY MANAGEMENT'.
2. Select 'CONFIGURE' and provide the Broadcom entitlement/download token.
3. Enter the token in the Download Token field. In this lab environment, the value is supplied automatically.
4. Select 'AUTHENTICATE' to validate the token and save the depot configuration.

Once authentication succeeds, the online depot is ready to use.



Download the Required Software

Download the software components that will be used during the VCF deployment.

1. Scroll through the Binary Management section to display the available components.
2. For this lab, select the components associated with VCF 9.0.0 by using the selection box at the top of the list.
3. Select 'DOWNLOAD' to retrieve the selected binaries from the configured depot.

The installer downloads and validates each component. A successful download is indicated by the 'Success' status.

VMware Cloud Foundation Installer

Hello, admin@local

EDIT DEPOT CONNECTIONDISCONNECT

CONFIGURE

Binary Management

Download Summary

Version: 9.0.0.0VMware vSphere FoundationNot downloadedVMware Cloud FoundationNot downloaded

Filter the list by product and version and download the required component binaries. For additional guidance, refer to the [Product Interoperability Matrix](#)

Product VMware Cloud FoundationVersion 9.0.0

DOWNLOADDELETE

☒	Component	Version	Size	Download Status
☒	SDDC Manager	9.0.0.0	2.00 GB	Not downloaded
☒	VMware Cloud Foundation Automation	9.0.0.0	21.17 GB	Not downloaded
☒	VMware Cloud Foundation Operations	9.0.0.0	2.63 GB	Not downloaded
☒	VMware Cloud Foundation Operations Collector	9.0.0.0	2.68 GB	Not downloaded
☒	VMware Cloud Foundation Operations fleet management	9.0.0.0	1.49 GB	Not downloaded
☒	VMware NSX	9.0.0.0	9.99 GB	Not downloaded
☒	VMware vCenter	9.0.0.0	12.04 GB	Not downloaded

VMware Cloud Foundation Installer

Hello, admin@local

EDIT DEPOT CONNECTIONDISCONNECT

CONFIGURE

Binary Management

Download Summary

Version: 9.0.0.0VMware vSphere FoundationDownloadedVMware Cloud FoundationDownloaded

Filter the list by product and version and download the required component binaries. For additional guidance, refer to the [Product Interoperability Matrix](#)

Product VMware Cloud FoundationVersion 9.0.0

DOWNLOADDELETE

☐	Component	Version	Size	Download Status
☐	SDDC Manager	9.0.0.0	2.00 GB	Success
☐	VMware Cloud Foundation Automation	9.0.0.0	21.17 GB	Success
☐	VMware Cloud Foundation Operations	9.0.0.0	2.63 GB	Success
☐	VMware Cloud Foundation Operations Collector	9.0.0.0	2.68 GB	Success
☐	VMware Cloud Foundation Operations fleet management	9.0.0.0	1.49 GB	Success
☐	VMware NSX	9.0.0.0	9.99 GB	Success
☐	VMware vCenter	9.0.0.0	12.04 GB	Success

1. Return to the Installer Home Page
2. Scroll back to the top of the page and select 'RETURN HOME' to continue with the deployment workflow.

VMware Cloud Foundation Installer Hello, admin@local

[RETURN HOME](#)

Depot Settings

You can only connect the VCF Installer to one depot at a time. Configuring another depot connection deactivates and deletes any existing configuration.

Alternatively, you can use the Binary Transfer utility to manually download the binaries from the depot to your local computer and then upload them to the VMware Cloud Foundation Installer appliance. [Read more](#)

Connect to the online depot

Depot connection active
Proxy settings are not configured

[EDIT DEPOT CONNECTION](#) [DISCONNECT](#)

Offline Depot

Offline depot connection not set up. Start setup and enter the offline depot settings to setup connection.

[CONFIGURE](#)

Binary Management

Download Summary

Version: 9.0.0.0	VMware vSphere Foundation	VMware Cloud Foundation
	Downloaded	Downloaded

Filter the list by product and version and download the required component binaries. For additional guidance, refer to the [Product Interoperability Matrix](#)

Product: VMware Cloud Foundation Version: 9.0.0

[DOWNLOAD](#) [DELETE](#)

<https://10.1.10.250/vcf-installer-ui/portal>

Start the VCF Deployment

The VCF deployment can now be initiated from the VCF Installer workflow.

1. Select 'DEPLOYMENT WIZARD' to open the deployment workflow.
2. From the platform selection menu, choose 'VMware Cloud Foundation'.

VMware Cloud Foundation Installer Hello, admin@local

VMware Cloud Foundation Installer

VMware Cloud Foundation installer deploys and performs the initial configuration of a new VMware vSphere Foundation or VMware Cloud Foundation environment. The installation process automates the deployment and configuration of the various components in each solution.

[LEARN MORE](#) [REVIEW PREREQUISITES](#)



Get Started

Download Binaries

Online depot connection active

You can edit your depot connection and/or download additional software binaries

VMware vSphere Foundation 9.0.0.0 Downloaded

VMware Cloud Foundation 9.0.0.0 Downloaded

[DEPOT SETTINGS](#)

Deploy

You can deploy VMware Cloud Foundation or VMware vSphere Foundation using the deployment wizard or uploading a deployment specification JSON file.

[DEPLOYMENT WIZARD](#) [DEPLOY USING JSON SPEC](#)

VMware Cloud Foundation

VMware vSphere Foundation

For this lab, a new VCF Fleet is created. The installer also supports adding a new VCF Instance to an existing fleet when that architecture is already available.

1. Scroll to the bottom of the page to review the available deployment options.
2. Select 'CONTINUE' to start the configuration process.

VMware Cloud Foundation Installer | Hello, admin@local

Deploy VMware Cloud Foundation: What would you like to deploy?

[LEARN ABOUT VCF FLEET](#)

☒ Deploy a new VCF fleet

☐ Deploy a VCF Instance in an existing VCF fleet

VCF Fleet

VCF Operations

VCF Automation

First VCF Instance

Management Domain

vCenter

NSX Manager

vSphere Cluster

SDDC Manager

Workload Domain(s)

Additional VCF Instance(s)

Existing VCF Fleet

VCF Operations

VCF Automation

Existing VCF Instance(s)

Additional VCF Instance(s)

Management Domain

vCenter

NSX Manager

vSphere Cluster

SDDC Manager

Workload Domain(s)

① If you intend to reuse existing components like a VMware vCenter or VCF Operations instance, subsequent steps will guide you to use them as building blocks in your deployment.

[CANCEL](#) [CONTINUE](#)

Existing Components

The installer can use existing VCF Operations or vCenter components when required. In this implementation, these services are deployed as part of the new environment rather than reused.

1. Select 'NEXT' to continue.

VMware Cloud Foundation Installer | Hello, admin@local

Deploy VCF Fleet

Existing Components

1 Existing Components

2 General Information

3 VCF Operations

4 VCF Automation

5 vCenter

6 NSX Manager

7 Storage

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Existing Components

Select the existing components you want to reuse in this VCF fleet. You will be asked to provide details about the existing components later in the deployment wizard.

- VCF Installer validates the selected components to ensure that they are compatible.
- If you select a VCF Operations instance that is connected to a fleet management appliance or a VCF Automation instance, these components will also be added to this VCF fleet.
- It is recommended you select the VMware vCenter where the selected VCF Operations is currently deployed to be used as the vCenter for the first VCF instance.
- If a VMware NSX Manager is connected to the vCenter instance you intend to use, select the option to use existing NSX Manager.

☐ **VCF Operations**

Use an existing VCF Operations instance as the management component in the VCF fleet.

☐ **VMware vCenter**

Use an existing vCenter as the management domain vCenter.

VCF Fleet

VCF Operations

VCF Automation

VCF Instance

Management Domain

vCenter

NSX Manager

vSphere Cluster

SDDC Manager

[CLOSE](#) [NEXT](#)

Deployment Parameters

Enter the basic identity and deployment parameters for the VCF environment. The lab uses VCF 9.0.0 and a simple single-node appliance model.

1. Select the VCF Instance Name field.
2. Enter 'vcf-a'.
3. Select the Management Domain Name field.
4. Enter 'mgmt-a'.
5. Choose 'Simple (Single-node)' for the appliance deployment model used in this lab.
6. Enter the DNS domain associated with the VCF environment.
7. Use 'site-a.vcf.lab'.
8. Specify the DNS server used by the deployment.
9. Enter '10.1.1.1'.

VMware Cloud Foundation Installer | Hello, admin@local

Deploy VCF Fleet

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General Information

Enter the general configuration details for this deployment. All fields marked with a * are required.

Version * 9.0.0.0
The latest downloaded version of each component will be used

VCF Instance Name * vcf-a
Descriptive name for the VCF instance

Management domain name * mgmt-a
This name will also be used to generate names for objects in the management domain, like the vCenter, distributed switches etc.

Advanced: Custom networking configuration for VCF Operations and VCF Automation
☐ I want to connect VCF Operations and VCF Automation instances on different DVPGs or NSX segments
By default, the VCF installer configures all VCF Operations and VCF Automation appliances to use the same distributed virtual port group (DVPG). To specify a different configuration, select the checkbox and perform a custom deployment of VCF Operations and VCF Automation after VCF installer deploys the other components. [Learn more](#)

Deployment model *
This selection only applies to newly deployed appliances of VCF Operations, VCF Automation and NSX Manager. Existing appliances that you plan to use in your VCF will not be modified. [Learn more](#)

☒ Simple (Single-node)
☐ High Availability (Three-node)

DNS and NTP servers
This selection only applies to newly deployed appliances.

DNS domain name * site-a.vcf.lab

DNS servers * 10.1.1.1
Enter multiple DNS servers as comma-separated.

VCF Fleet

- VCF Operations
- VCF Automation

VCF Instance

- Management Domain**
 - vCenter
 - NSX Manager
 - vSphere Cluster
 - SDDC Manager

CLOSE BACK NEXT

10. Scroll down to expose the remaining general configuration fields.
11. Enter the NTP server address.
12. Use '10.1.1.1' for time synchronization in this lab.
13. For this implementation, enable 'Auto-generate passwords for newly installed appliances' so the installer creates credentials for the deployed components.
14. Select 'NEXT' to continue.

VMware Cloud Foundation Installer

Hello, admin@local

Deploy VCF Fleet

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4 VCF Automation

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General Information

Advanced: Custom networking configuration for VCF Operations and VCF Automation

☐ I want to connect VCF Operations and VCF Automation instances on different DVPs or NSX segments

By default, the VCF Installer configures all VCF Operations and VCF Automation appliances to use the same distributed virtual port group (DVP). To specify a different configuration, select the checkbox and perform a custom deployment of VCF Operations and VCF Automation after VCF Installer deploys the other components. [Learn more](#)

Deployment model *

This selection only applies to newly deployed appliances of VCF Operations, VCF Automation and NSX Manager. Existing appliances that you plan to use in your VCF will not be modified. [Learn more](#)

☒ Simple (Single-node)

☐ High Availability (Three-node)

DNS and NTP servers

This selection only applies to newly deployed appliances.

DNS domain name

DNS servers *
Enter multiple DNS servers as comma-separated.

NTP servers *
Enter multiple NTP servers as comma-separated.

Password creation

☒ Auto-generate passwords for newly installed appliances

Other

☒ Enable Customer Experience Improvement Program (CEIP)

BACK

NEXT

vCenter

NSX Manager

vSphere Cluster

SDDC Manager

CLOSE

VCF Operations Configuration

Define the parameters for the VCF Operations services. Because the deployment model is single-node, only the required single-node settings are presented.

1. Open the Operations Appliance Size selector.
2. Select 'Small' for the lab deployment.
3. Enter the primary Operations FQDN.
4. Use 'ops-a.site-a.vcf.lab'.
5. Specify the Fleet Management Appliance FQDN. This appliance provides fleet-level deployment and lifecycle management functions.
6. Use 'opslcm-a.site-a.vcf.lab'.
7. Enter the Operations Collector Appliance FQDN.
8. Use 'opscollector-01a.site-a.vcf.lab'.
9. Select 'NEXT' to continue.

VMware Cloud Foundation Installer

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VCF Operations

Enter the configuration details for the VCF Operations appliances. All fields marked with a * are required.

VCF Operations Appliance

Operations Appliance Size *
Small

4 vCPUs and 16GB Memory

Operations primary FQDN *
ops-a.site-a.vcf.lab

Fleet Management Appliance

Appliance FQDN *
opsfcm-a.site-a.vcf.lab

Operations Collector Appliance

Provide details for deploying a new operations collector appliance for this VCF instance.

Appliance FQDN *
opscollector-01a.site-a.vcf.lab

VCF Fleet

VCF Operations

VCF Automation

VCF Instance

Management Domain

vCenter

NSX Manager

vSphere Cluster

SDDC Manager

CLOSE

BACK

NEXT

VCF Automation Configuration

Configure the VCF Automation service. The lab deploys a new Automation appliance rather than connecting to an existing instance.

1. Enter the Automation Appliance FQDN.
2. Use 'auto-a.site-a.vcf.lab'.
3. Specify the first Automation node IP address.
4. Use '10.1.1.71'.
5. Specify the second node IP address.
6. Use '10.1.1.72'.
7. Define the node naming prefix.
8. Use 'auto-a'.
9. Select 'NEXT' to continue.

VMware Cloud Foundation Installer

Hello, admin@local

Deploy VCF Fleet

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VCF Automation

Enter the configuration details for VCF Automation appliance. All fields marked with a * are required.

☐ I want to connect a VCF Automation instance later
Choose this option if you wish to use an existing Aria Automation instance that is not currently connected to the VCF Operations instance you specified in the previous step. By doing so, you agree to use VCF Operations to import your Aria Automation instance once the installer is complete.

Appliance FQDN *

auto-a.site-a.vcf.lab

VCF Automation requires single node in a simple deployment. In addition provide second address to be used when recreating a node for upgrade

Node IP 1 *

10.11.71

Node IP 2 *

10.11.72

Node name prefix ⓘ *

auto-a

Internal Cluster CIDR ⓘ *

198.18.0.0/15

VCF Fleet

☒ VCF Operations

VCF Automation

VCF Instance

Management Domain

vCenter

NSX Manager

vSphere Cluster

SDDC Manager

CLOSE

BACK

NEXT

vCenter Configuration

Define the vCenter appliance that will manage the initial Management Domain.

1. Enter the vCenter Appliance FQDN.
2. Use 'vc-mgmt-a.site-a.vcf.lab'.

The remaining values can remain at their generated defaults for this lab. Review the available fields before proceeding.

1. Select 'NEXT' to continue.

VMware Cloud Foundation Installer

Hello, admin@local

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vCenter

Enter the configuration details for the management domain vCenter. All fields marked with a * are required.

Appliance FQDN *

vc-mgmt-a.site-a.vcf.lab

Appliance Size *

Small

Appliance Storage Size

4 vCPUs and 21 GB of memory

Default

694 GB of disk space

Datacenter Name *

mgmt-a-dc01

Cluster Name *

mgmt-a-ci01

SSO Domain Name

vsphere.local

VCF Fleet

☒ VCF Operations

☒ VCF Automation

VCF Instance

Management Domain

vCenter

NSX Manager

vSphere Cluster

SDDC Manager

CLOSE

BACK

NEXT

NSX Manager Configuration

Configure the NSX Manager instance associated with the Management Domain. The lab uses the default Medium appliance size for the single-node deployment.

1. Enter the NSX Manager cluster FQDN.
2. Use 'nsx-mgmt-a.site-a.vcf.lab'.
3. Enter the NSX Manager appliance FQDN.
4. Use 'nsx-mgmt-01a.site-a.vcf.lab'.
5. Select 'NEXT' to continue.

The screenshot shows the 'NSX Manager' configuration step in the VMware Cloud Foundation Installer. The left sidebar lists the deployment steps, with '6 NSX Manager' selected. The main area is titled 'NSX Manager' and contains the following fields:

- Appliance Size ***: Medium (dropdown menu)
- Cluster FQDN ***: nsx-mgmt-a.site-a.vcf.lab
- Appliance FQDN ***: nsx-mgmt-01a.site-a.vcf.lab

Below these fields, there is a 'VCF Instance' section with a 'Management Domain' box containing:

- ☒ vCenter
- ☐ NSX Manager
- ☐ vSphere Cluster
- ☐ SDDC Manager

At the bottom right, there are buttons for 'CLOSE', 'BACK', and 'NEXT'.

Management Domain Storage

Select the storage technology that will back the Management Domain. The available choices in this workflow include vSAN, VMFS on Fibre Channel, and NFS v3.

1. Open the vSAN Architecture selector.
2. For this implementation, select 'vSAN OSA'.

Keep the remaining storage values at their generated defaults and review the available configuration options.

1. Select 'NEXT' to continue.

Management Domain Hosts

Add the four ESXi hosts that will form the initial Management Domain cluster.

1. Enter the ESXi root password in the ESX Root Password field.
2. Use 'ADD HOST' until four host entries are available.
3. Enter the FQDN for the first ESXi host.
4. Use 'esx-01a.site-a.vcf.lab'.
5. Select 'ADD' to register the host.
6. Enter the FQDN for the second host.
7. Use 'esx-02a.site-a.vcf.lab'.
8. Select 'ADD'.
9. Enter the FQDN for the third host.
10. Use 'esx-03a.site-a.vcf.lab'.
11. Select 'ADD'.
12. Enter the FQDN for the fourth host.
13. Use 'esx-04a.site-a.vcf.lab'.
14. Select 'ADD' to register the final host.
15. Select 'CONFIRM ALL FINGERPRINTS' and verify the SSL fingerprints presented for the hosts.
16. Select 'NEXT' to continue.

VMware Cloud Foundation Installer | Hello, admin@local

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Hosts

Select the hosts to add to the management domain of your VCF Instance. All fields marked with a * are required.

ESX Root Password *

Add Hosts

The minimum number of required hosts is displayed in the table. Click Add Host to add more ESX hosts.

	FQDN	Confirm Fingerprint
☐	esx-01a.site-a.vcf.lab	36:B6:F9:9E:E6:6A:AE:88:17:E6:90:34:C9:37:97:DA:D4:2B:32:BA:CB:F3:06:F7:31:F9:6B:25:C6:C8:94:9D
☐	esx-02a.site-a.vcf.lab	87:DD:78:5F:0A:BA:6B:8F:44:D8:E3:D1:79:81:88:11:E9:88:3A:FD:FD:CD:DB:F2:69:8C:16:75:2E:10:AD:DD
☐	esx-03a.site-a.vcf.lab	A8:C6:08:3F:64:F2:0B:01:CB:B3:D3:A2:B7:16:87:7A:AA:4C:4C:A9:26:8F:7F:BD:FF:80:0D:0D:BE:71:20:42
☐	esx-04a.site-a.vcf.lab	D1:DD:7B:E2:9D:6F:AC:B7:FA:DA:58:AC:88:25:AC:0E:37:59:07:14:CD:80:C1:C5:A3:37:47:AF:14:AA:AA:50

VCF Fleet

- ☒ VCF Operations
- ☒ VCF Automation

VCF Instance

Management Domain

- ☒ vCenter
- ☒ NSX Manager
- ☒ vSphere Cluster
- ☐ SDDC Manager

Network Addressing and Subnetting

Before entering the network values in the installer, define the IP addressing plan and subnet boundaries. The design separates management, vMotion, vSAN, and NSX overlay traffic so that each traffic type has its own broadcast domain and addressing range.

Function	VLAN	Subnet	Gateway	Usable Hosts	Broadcast
Management	10	10.1.1.0/24	10.1.1.1	10.1.1.1–10.1.1.254	10.1.1.255
vSAN	11	10.1.2.0/25	10.1.2.1	10.1.2.1–10.1.2.126	10.1.2.127
vMotion	13	10.1.3.0/25	10.1.3.1	10.1.3.1–10.1.3.126	10.1.3.127
NSX Overlay / TEP*	14	10.1.3.128/25*	10.1.3.129*	10.1.3.129–10.1.3.254*	10.1.3.255*

* NSX Overlay subnet shown according to the network plan in the provided notes; the installer workflow uses DHCP for TEP assignment.

Management Network – VLAN 10

1. The management network uses VLAN 10.
2. VLAN ID: 10.
3. Subnet: 10.1.1.0/24.

4. A /24 mask provides 256 addresses in total, with 254 usable host addresses. The network address is 10.1.1.0, the gateway is 10.1.1.1, and the broadcast address is 10.1.1.255.
5. Default Gateway.
6. Use 10.1.1.1 as the management gateway.

VM Management Network

1. Use the same addressing information as the ESXi Management Network for VM Management.

vMotion Network – VLAN 13

1. The vMotion segment uses VLAN 13.
2. VLAN ID: 13.
3. Subnet: 10.1.3.0/25.
4. A /25 mask divides the /24 into two 128-address subnets. For vMotion, 10.1.3.0/25 provides addresses from 10.1.3.0 through 10.1.3.127, with 126 usable host addresses.
5. Set the MTU to 8000 for the vMotion traffic path.
6. MTU: 8000.
7. Default Gateway.
8. Use 10.1.3.1 as the vMotion gateway.
9. IP Address Pool.
10. Reserve the range 10.1.3.101 through 10.1.3.120 for vMotion hosts. The range stays within the usable portion of the 10.1.3.0/25 subnet.
11. Pool End Address.
12. Set the upper limit to 10.1.3.120.

vSAN Network – VLAN 11

1. The vSAN segment uses VLAN 11.
2. VLAN ID: 11.
3. Set the vSAN MTU to 8000.
4. MTU: 8000.
5. Subnet: 10.1.2.0/25.
6. The 10.1.2.0/25 subnet contains 128 addresses: 10.1.2.0 is the network address, 10.1.2.1 is the gateway, 10.1.2.127 is the broadcast address, and 10.1.2.1–10.1.2.126 are usable host addresses.
7. Default Gateway.

8. Use 10.1.2.1 as the vSAN gateway.
9. Scroll down to the address pool fields.
10. IP Address Pool Start.
11. Use 10.1.2.101 as the beginning of the vSAN address pool.
12. Pool End Address.
13. Use 10.1.2.120 as the end of the vSAN address pool.
14. Select 'NEXT' to continue.

VMware Cloud Foundation Installer

Hello, admin@local

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Networks

physical switches in your environment. All fields marked with a * are required.

ESX Management Network

VLAN ID * 10

CIDR Notation * 10.1.0/24

MTU * 1500

Gateway * 10.1.1

VM Management Network

☒ Use same inputs from ESX Management Network

vMotion Network

VLAN ID * 13

CIDR Notation * 10.1.3.0/25

IP Address Range * 10.1.3.101

MTU * 8000

Gateway * 10.1.3.1

To * 10.1.3.120

vSAN Network

VLAN ID * 11

CIDR Notation * 10.1.2.0/25

IP Address Range * 10.1.2.101

MTU * 8000

Gateway * 10.1.2.1

To * 10.1.2.120

VCF Instance

Management Domain

- ☒ vCenter
- ☒ NSX Manager
- ☒ vSphere Cluster
- ☐ SDDC Manager

CLOSE BACK **NEXT**

Virtual Distributed Switch and NSX Traffic Separation

Configure the Management Domain distributed switching design. Each ESXi host has four physical NICs, allowing the configuration to dedicate a pair of uplinks to NSX traffic while the remaining uplinks carry the other infrastructure traffic.

1. Scroll down to the distributed switch configuration.
2. Enable 'NSX Traffic Separation' to create separate distributed switches for NSX and non-NSX traffic.
3. Locate the MTU setting for 'mgmt-a-cl01-vds02'.
4. Set the MTU to 8000.

VMware Cloud Foundation Installer Hello, admin@local

Deploy VCF Fleet

- 1 Existing Components
- 2 General Information
- 3 VCF Operations
- 4 VCF Automation
- 5 vCenter
- 6 NSX Manager
- 7 Storage
- 8 Hosts
- 9 Networks
- 10 Distributed Switch**
- 11 SDDC Manager
- 12 Review
- 13 Validate & Deploy

Distributed Switch

Provide the vSphere Distributed Switch configuration to be applied to the hosts in the cluster.

4 physical network adapters are available to use. [View Details](#)

Selected Profile: NSX Traffic Separation [CHANGE](#)

This profile creates two distributed virtual switches with separate physical NICs. One switch is dedicated for NSX Edge and overlay traffic while the other is used for all other traffic.

Distributed Switches [RESET CHANGES](#)

Distributed Switch Name
mgmt-a-cl01-vds01
mgmt-a-cl01-vds02

mgmt-a-cl01-vds02

Distributed Switch Name: mgmt-a-cl01-vds02

MTU: 8000

Number of Uplinks: 2

Map uplinks to physical network adapters on host:

uplink1	vmnic2
uplink2	vmnic3

Network Traffic: NSX

☒ Apply default operation mode configured in NSX Manager

VCF Fleet

- VCF Operations
- VCF Automation

VCF Instance

Management Domain

- vCenter
- NSX Manager
- vSphere Cluster
- SDDC Manager

CLOSE BACK NEXT

- Continue to the NSX traffic settings.
- Configure the NSX Overlay VLAN.
- Use VLAN 14 for the NSX overlay segment.
- For this lab, select 'DHCP' for the TEP IP assignment method.
- Select 'NEXT' to continue.

VMware Cloud Foundation Installer Hello, admin@local

Deploy VCF Fleet

- 1 Existing Components
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Distributed Switch

Number of Uplinks: 2

Map uplinks to physical network adapters on host:

uplink1	vmnic2
uplink2	vmnic3

Network Traffic: NSX

☒ Apply default operation mode configured in NSX Manager

Load Balancing: Route based on the source of the port ID

uplink1: ☒ Active ☐ Standby ☐ Unused

uplink2: ☒ Active ☐ Standby ☐ Unused

Transport Zones

NSX-Overlay

Transport Zone Name: VCF-Created-Overlay-Zone

VLAN ID: 14

IP Assignment (TEP): ☒ DHCP ☐ IP Pool

CLOSE BACK NEXT

SDDC Manager

Specify the FQDN for the SDDC Manager appliance that will orchestrate and manage the VCF Management Domain.

1. Enter the SDDC Manager Appliance FQDN.
2. Use 'sddcmanager-a.site-a.vcf.lab'.
3. Select 'NEXT' to continue.

The screenshot shows the 'SDDC Manager' configuration step in the VMware Cloud Foundation Installer. The left sidebar lists the deployment steps, with '11 SDDC Manager' selected. The main area is titled 'SDDC Manager' and contains a message: 'Enter the configuration details for the SDDC Manager appliance. All fields marked with a * are required. The VMware Cloud Foundation installer appliance is not deployed on one of the hosts in the management domain. During the deployment process, a new SDDC Manager appliance will be deployed.' Below this, the 'Appliance FQDN *' field is populated with 'sddcmanager-a.site-a.vcf.lab'. A note below the field states: 'Either a Fully Qualified Domain name e.g host.vsphere.local or just the hostname e.g. host'. On the right, the 'VCF Fleet' section shows 'VCF Operations' and 'VCF Automation' as checked. The 'VCF Instance' section shows the 'Management Domain' with 'vCenter', 'NSX Manager', 'vSphere Cluster', and 'SDDC Manager' all checked. At the bottom right, there are 'CLOSE', 'BACK', and 'NEXT' buttons, with 'NEXT' being highlighted.

Configuration Review

Review the complete configuration before starting the validation and deployment process. Confirm the names, FQDNs, network ranges, VLANs, storage settings, host information, and appliance parameters.

1. Scroll through the summary to inspect the remaining configuration items.
2. Select 'NEXT' after confirming the configuration.

VMware Cloud Foundation Installer

Hello, admin@local

Deploy VCF Fleet

1 Existing Components

2 General Information

3 VCF Operations

4 VCF Automation

5 vCenter

6 NSX Manager

7 Storage

8 Hosts

9 Networks

10 Distributed Switch

11 SDDC Manager

12 Review

13 Validate & Deploy

Review

▼ Distributed Switch	mgmt-a-cl01-vds02
MTU	8000
▼ Uplinks mappings	
vmnic2	uplink1
vmnic3	uplink2
▼ Network traffic	
▼ NSX	
Operational Mode	Apply default operation mode configured in NSX Manager
Load Balancing	Route based on the source of the port ID
Active Uplinks	uplink1, uplink2
▼ Transport Zone Name	VCF-Created-Overlay-Zone
VLAN ID	14
IP Assignment (TEP)	DHCP
▼ SDDC Manager	
Appliance FQDN	sddcmanager-a.site-a.vcf.lab
VCF Installer Deployment	New SDDC Manager Deployment

CLOSE

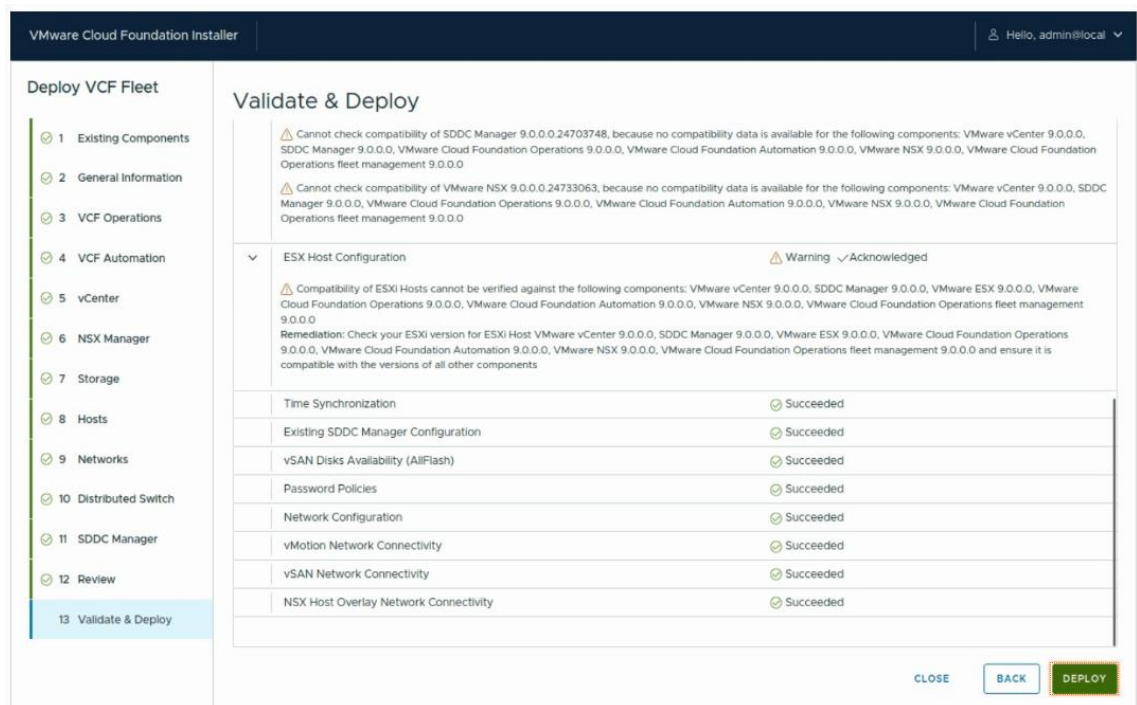
BACK

NEXT

Validate and Deploy

The VCF Installer performs automated checks against the configuration before deployment. In a simulated lab, some warnings may be expected because of the characteristics of the environment.

1. Select 'ACKNOWLEDGE ALL WARNINGS' when the warnings are known and acceptable. Resolve any actual validation errors before continuing.
2. Scroll down to the deployment controls.
3. Select 'DEPLOY' to start the VCF deployment.



Deployment Completed

After the deployment finishes, review the task results to confirm that the VCF components were created successfully.

1. Expand the task groups to inspect the individual operations performed by the installer. The workflow automates the configuration of the VCF components based on the values entered during the previous steps.
2. Collapse the task groups when the review is complete.
3. Scroll back toward the top of the completion page.
4. Because automatic password generation was selected, use 'REVIEW PASSWORDS' to display the credentials generated for the deployed components.

VMware Cloud Foundation Installer

Hello, admin@local

[RETURN HOME](#)



Congratulations! Your deployment completed successfully!

You can now follow the next steps recommendations described in the right panel

[DOWNLOAD JSON SPEC](#) [REVIEW PASSWORDS](#)

100% completed

- ✓ Deploy vCenter
- ✓ Deploy SDDC Manager
- ✓ Configure the vSphere cluster
- ✓ Deploy and configure NSX



Next Steps

1. Log in to the VCF Operations UI.
2. Navigate to License Management, and apply your VCF license keys within the 60-day evaluation period.

VCF Operations Login

[ops-a-site-a-vcf-lab](#) ✓

Username: admin

Password: *****

Tasks

VMware Cloud Foundation Deployment

[REFRESH](#)

Subtask	Description	Status	Start time	Last updated
Record VCF Installer Workflow Execution completion in Customer Experience Improvement Program (CEIP)	Record VCF Installer Workflow Execution completion in Customer Experience Improvement Program (CEIP)	Success	5/27/25, 3:25 AM	5/27/25, 3:25 AM
Updating deployment status of VCF Automation to succeeded	Updating deployment status of VCF Automation to succeeded	Success	5/27/25, 3:25 AM	5/27/25, 3:25 AM

The completion screen provides the FQDNs and credentials associated with the deployed appliances. Password changes can subsequently be performed through VCF Operations as part of day-2 administration.

The VCF Private Cloud deployment procedure is complete.